Feiyu Lu, PhD

Research Scientist

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About Me

Feiyu Lu is a Research Scientist in the Immersive Technology & Spatial Computing Team, Global Technology Applied Research Group at JPMorgan-Chase. He obtained his Ph.D. degree at Virginia Tech advised by Prof. Doug A. Bowman. His research interest lies in the intersections of **Human-AI Interaction**, **Spatial/Multimodal Interaction**, **Artificial Intelligence / Machine Learning**, and **Augmented/Virtual/Mixed Reality**.

Education

Virginia Polytechnic Institute and State University (Virginia Tech) — *Blacksburg, VA, USA*

Ph.D. in Computer Science & Applications — *Aug 2018 - May 2023*

- Research Group: **3D Interaction Group**
- Advisor: **Prof. Doug A. Bowman**
- Dissertation: **Glanceable AR - Towards a Pervasive and Always-On Augmented Reality Future**
- Committee Members: Prof. Joseph L. Gabbard, Prof. Wallace Lages, Prof. Sang Won Lee, Prof. Yalong Yang, Prof. Steven K. Feiner

Xi'an Jiaotong-Liverpool University (XJTLU) — *Suzhou, Jiangsu, China*

B.Eng in Computer Engineering — *Sep 2014 - Jun 2018*

- Research Group: X-CHI Group
- Advisor: **Prof. Hai-Ning Liang**
- Graduated with First-Class Honor (Top 5%)

Research Experience

Immersive Technology & Spatial Computing Team, JPMorganChase — *New York, NY, USA*

Research Scientist — *May 2023 - Current*

- I am working as an Applied Research Scientist in the Global Technology Applied Research Group at JPMC.
- I investigate novel integrations of immersive technology, interactive wall displays, and AI/ML techniques such as reinforcement learning, large language models, and large multimodal models to addressing concrete business problems and advancing our clients & employee experiences.
- I work with internal stakeholders, build novel intelligent systems, conduct user-centric research, and deliver tech-transfer-ready proof-of-concept systems to boost employee productivity and client engagement.
- I have contributed to six patents, five full papers, and three extended abstracts at top conferences.
- Intern Mentored: **Dr. Siyou Pei**
- Manager: **Prof. Blair MacIntyre**

Reality Labs Research, Meta Inc. — *Redmond, WA, USA*

Research Scientist Intern — *May 2022 - Aug 2022*

- I worked with teams at Reality Labs Research to perform user experience research on AR/VR interactions.
- I researched how user and system factors may influence the perceived human-AI interaction experience in AR/VR.
- I conducted a remote VR study with over 80 participants.
- The work was published and presented at ISMAR'23.
- Mentors: Yan Xu, Brennan Jones, Missie Smith, Laird Malamed & Sophie Kim

Reality Labs Research, Meta Inc. — *Remote*

Research Intern — *May 2021 - Aug 2021*

- I worked with teams at Reality Labs Research to perform usability evaluations on intelligent AR interfaces.
- I conducted workshops with UX researchers and designers to identify pain-points during AR interactions.
- I conducted a remote VR study with over 40 participants.
- The work was patented and published at CHI'22.
- Mentors: Yan Xu, Missie Smith & Sophie Kim

3D Interaction Group — *Blacksburg, VA, USA*

Graduate Researcher — *Aug 2018 - May 2023*

- I worked as a GRA in the 3D Interaction Group directed by Prof. Doug A. Bowman.
- I worked on exploring interaction techniques for lightweight activation of everyday AR content.
- I worked on designing, implementing and evaluating general-purpose systems for AR HWDs.
- I worked on developing efficient and unobtrusive information access strategies for AR HWDs.
- I worked on evaluating gaze direction visualization techniques in wide-area collaborative AR environments.
- Advisor: Prof. Doug A. Bowman

Selected Publications

I published the majority of my work in premiere conferences / journals on spatial user interactions / human-computer interactions / immersive technologies, including IEEEVR, ISMAR, TVCG and ACM CHI, SUI. The typical acceptance rates of these venues are around 25%. Below is a selected list of my publications. The full paper list is attached to the end of this resume.

Journal Articles

Revisiting Performance Models of Distal Pointing Tasks in Virtual Reality
Logan Lane, **Feiyu Lu**, Shakiba Davari, Robert J. Teather, Doug A. Bowman
IEEE Transactions on Visualization and Computer Graphics. 2025
DOI: 10.1109/TVCG.2025.3567078

"Where Did My Apps Go?" Supporting Scalable and Transition-Aware Access to Everyday Applications in Head-Worn Augmented Reality
Lu, Feiyu, Leonardo Pavanatto, Shakiba Davari, Lei Zhang, Lee Lisle, Doug A. Bowman
IEEE Transactions on Visualization and Computer Graphics. 2025
DOI: 10.1109/TVCG.2024.3493115

Multiple Monitors or Single Canvas? Evaluating Window Management and Layout Strategies on Virtual Displays
Leonardo Pavanatto, **Feiyu Lu**, Chris North, Doug A. Bowman
IEEE Transactions on Visualization and Computer Graphics (TVCG). 2024
DOI: 10.1109/TVCG.2024.3368930

Conference Proceedings

S-TIER: Situated-Traceable Insights in Extended Reality for Hybrid Crisis Management
Feiyu Lu*, Cheng Yao Wang*, Leonardo Pavanatto, Fannie Liu, David Saffo, Benjamin Lee, Mengyu Chen, Blair MacIntyre
In submission to 2026 IEEE International Symposium on Mixed and Augmented Reality (ISMAR), 2026, Bari, Italy
DOI: InSubmission

Anu.js: Accelerating Web-based Immersive Analytics
David Saffo*, Benjamin Lee*, **Feiyu Lu**, Cheng Yao Wang, Blair MacIntyre
Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems, 2026, Creant el demà junts, Barcelona
DOI: Accepted

Evaluating the Viability of Additive Models to Predict Task Completion Time for 3D Interactions in Augmented Reality
Logan Lane, Ibrahim Tahmid, **Feiyu Lu**, Doug A Bowman
2026 IEEE Virtual Reality and 3D User Interfaces (VR), 2026, Daegu, South Korea
DOI: Accepted

Examining the Effects of Immersive and Non-Immersive Presenter Modalities on Engagement and Social Interaction in Co-located Augmented Presentations
Matt Gottsacker, Mengyu Chen, David Saffo, **Feiyu Lu**, Benjamin Lee, Blair MacIntyre
Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems, 2025, Yokohama, Japan
DOI: 10.1145/3706598.3713346

SocialMiXR: Facilitating Hybrid Social Interactions at Conferences
Fannie Liu, Cheng Yao Wang, William Moriarty, **Feiyu Lu**, Usman Mir, David Saffo, Mengyu Chen, Blair MacIntyre
Proceedings of the ACM on Human-Computer Interaction (CSCW), 2025, Bergen, Norway
DOI: 10.1145/3711069

Exploring the Impact of User and System Factors on Human-AI Interactions in Head-Worn Displays
Feiyu Lu, Yan Xu, Xuhai Xu, Brennan Jones, Laird Malamed
2023 IEEE International Symposium on Mixed and Augmented Reality (ISMAR), 2023, Sydney, NSW, Australia
DOI: 10.1109/ISMAR59233.2023.00025

In-the-Wild Experiences with an Interactive Glanceable AR System for Everyday Use
Feiyu Lu, Leonardo Pavanatto, Doug A. Bowman
Proceedings of the 2023 ACM Symposium on Spatial User Interaction (SUI), 2023, Sydney, NSW, Australia
DOI: 10.1145/3607822.3614515

XAIR: A Framework of Explainable AI in Augmented Reality
Xuhai Xu, Anna Yu, Tanya R. Jonker, Kashyap Todi, **Feiyu Lu**, Xun Qian, João Marcelo Evangelista Belo, Tianyi Wang, Michelle Li, Aran Mun, Te-Yen Wu, Junxiao Shen, Ting Zhang, Narine Kokhlikyan, Fulton Wang, Paul Sorenson, Sophie Kim, Hrvoje Benko
Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems (CHI), 2023, Hamburg, Germany
DOI: 10.1145/3544548.3581500
Best Paper Honorable Mention

Exploring Spatial UI Transition Mechanisms with Head-Worn Augmented Reality
Feiyu Lu, Yan Xu
Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems (CHI), 2022, New Orleans, LA, USA
DOI: 10.1145/3491102.3517723

Validating the Benefits of Glanceable and Context-Aware Augmented Reality for Everyday Information Access Tasks
Shakiba Davari, **Feiyu Lu**, Doug A. Bowman
2022 IEEE Conference on Virtual Reality and 3D User Interfaces (VR), 2022, Christchurch, New Zealand
DOI: 10.1109/VR51125.2022.00063

Exploration of Techniques for Rapid Activation of Glanceable Information in Head-Worn Augmented Reality
Feiyu Lu, Shakiba Davari, Doug A. Bowman
Proceedings of the 2021 ACM Symposium on Spatial User Interaction (SUI), 2021, Virtual Event, USA
DOI: 10.1145/3485279.3485286
Best Paper Honorable Mention

Evaluating the Potential of Glanceable AR Interfaces for Authentic Everyday Uses
Feiyu Lu, Doug A. Bowman
2021 IEEE Virtual Reality and 3D User Interfaces (VR), 2021, Lisbon, Portugal
DOI: 10.1109/VR50410.2021.00104

Glanceable AR: Evaluating Information Access Methods for Head-Worn Augmented Reality
Feiyu Lu, Shakiba Davari, Lee Lisle, Yuan Li, Doug A. Bowman
2020 IEEE Conference on Virtual Reality and 3D User Interfaces (VR), 2020, Atlanta, GA, USA
DOI: 10.1109/VR46266.2020.00113

Gaze Direction Visualization Techniques for Collaborative Wide-Area Model-Free Augmented Reality
Yuan Li, **Feiyu Lu**, Wallace S Lages, Doug A. Bowman
Symposium on Spatial User Interaction (SUI), 2019, New Orleans, LA, USA
DOI: 10.1145/3357251.3357583

Evaluating Engagement Level and Analytical Support of Interactive Visualizations in Virtual Reality Environments
Feiyu Lu, Difeng Yu, Hai-Ning Liang, Wenjun Chen, Konstantinos Papangelis, Nazlena Mohamad Ali
2018 IEEE International Symposium on Mixed and Augmented Reality (ISMAR), 2018, Munich, Germany
DOI: 10.1109/ISMAR.2018.00050

Extended Abstracts

Adaptive 3D UI Placement in Mixed Reality Using Deep Reinforcement Learning
Feiyu, Lu*, Mengyu* Chen, Hsiang Hsu, Pranav Deshpande, Cheng Yao Wang, Blair MacIntyre
Extended Abstracts of the CHI Conference on Human Factors in Computing Systems, 2024, Honolulu, HI, USA
DOI: 10.1145/3613905.3651059

Clean the Ocean: An Immersive VR Experience Proposing New Modifications to Go-Go and WiM Techniques
Lee Lisle, **Feiyu Lu**, Shakiba Davari, Ibrahim Asadullah Tahmid, Alexander Giovannelli, Cory Llo, Leonardo Pavanatto, Lei Zhang, Luke Schlueter, Doug A. Bowman
2022 IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW), 2022, Christchurch, New Zealand
DOI: 10.1109/VRW55335.2022.00311
Best 3DUI Contest Entry Award

Fantastic Voyage 2021: Using Interactive VR Storytelling to Explain Targeted COVID-19 Vaccine Delivery to Antigen-presenting Cells
Lei Zhang, **Feiyu Lu**, Ibrahim Asadullah Tahmid, Shakiba Davari, Lee Lisle, Nicolas Gutkowski, Luke Schlueter, Doug A. Bowman
2021 IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW), 2021, Lisbon, Portugal
DOI: 10.1109/VRW52623.2021.00230
Best 3DUI Contest Entry Award

Achievements

2025	**Best Doctoral Dissertation Award**, IEEEVR'25	*Saint-Malo, France*
2024	**Prolific Inventor**, Global Technology Applied Research	*JPMorganChase*
2023	**Best Paper Honorable Mention Award (Top 5%)**, CHI'23	*Hamburg, Germany*
2022	**Best 3DUI Contest Entry Award (1 of 11 teams)**, IEEEVR'22	*Christchurch, New Zealand*
2021	**Best Paper Honorable Mention Award (Top 5%)**, SUI'21	*Virtual Event*
2021	**Finalist (Top 3.5% of 2,163 applicants)**, Facebook Ph.D. Fellowship	*Virtual Event*
2021	**Best 3DUI Contest Entry Award (1 of 17 teams)**, IEEEVR'21	*Lisbon, Portugal*

Patents

2025	**System and Method for Long-Term Video Analysis and Context-Aware Search Using Vision-Language Models**, No. 19/222,406	*with JPMorganChase*
2024	**Privacy-preserving Methods for Remote Space Sharing in Enterprise Environment**, No. 18/774,705	*with JPMorganChase*
2024	**Systems and Methods for Hybrid Cross-Reality Collaboration Around an Large Interactive Display**, No. 18/882,495	*with JPMorganChase*
2024	**HybridPortal: Enabling Hybrid Social Interactions for Hybrid Events**, No. 18/882,477	*with JPMorganChase*
2024	**Systems and Methods for Audience Feedback Guided Mixed Reality**, No. 18/769,647	*with JPMorganChase*
2023	**Systems and Methods for Placement of Virtual Content Objects in an Extended Reality Environment Based on Reinforcement Learning**, No. 18/407,003	*with JPMorganChase*
2021	**Dynamic Widget Placement Within an Artificial Reality Display**, No. 17/747,767	*with Meta Inc.*
2020	**Interacting with Glanceable Information in Wearable Augmented Reality**, No. 63/147,805	*with Virginia Tech*

Service

2025	**Program Committee - ISMAR**
2025	**Associate Chair (User Experience and Usability Track) - CHI**
2025	**Program Committee - IEEEVR**
2019	**Student Volunteer - IEEEVR**
2018	**Student Volunteer - ISMAR**

Reviewing (60^{+})

2026	**CHI *, IEEEVR**
2025	**CHI *, IEEEVR, ISMAR ***
2024	**CHI *, ISS *, MobileHCI, ISMAR**
2023	**CHI *, UIST *, ISS *, VRST**
2022	**IEEEVR, ISMAR *, SIGGRAPH, UIST**
2021	**CHI, IEEEVR, SIGGRAPH**

Invited Talks

2025	**Towards Pervasive Everyday Computing**, Hosted by Dr. Hai-Ning Liang	*The Hong Kong University of Science and Technology*
2024	**Immersive and Intelligent XR for Enterprise Use Cases**, Hosted by Dr. Zhen Bai	*University of Rochester, NY*
2024	**Industry Panelist - Intelligent XR Workshop**, Hosted by i-XR Workshop Organizers	*IEEE ISMAR, Seattle, WA*

Skills

Research	**3D Interaction, Human-AI Interaction, Multimodal Input, Spatial Computing, Extended Reality, Human-Computer Interaction.**
Programming/Platform	**Python, C#, JavaScript/TypeScript, Swift, WebXR, VisionOS, Meta Quest.**
Tools	**Unity3D, BabylonJS, Arduino, Blender, R, SPSS, Adobe Suites.**
Method	**Rapid Prototyping, Mixed-methods Research, Inferential Statistical, Thematic Analysis, Experimental Design, Usability Evaluation, Performance Modeling.**